NewsGenie: An AI-Powered Information and News Assistant

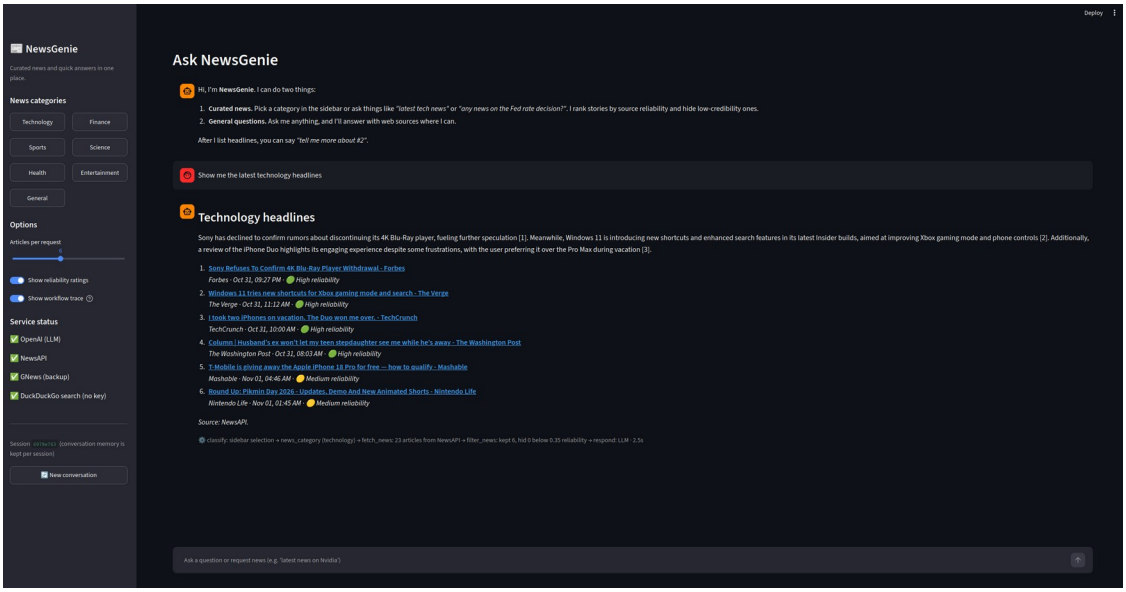
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Repository: github.com/Alessandrarye/newsgenie

1. Problem and approach

News consumption is fragmented. People bounce between apps for headlines, a search engine for background, and a chatbot for quick answers, and none of them tell you much about whether a source is trustworthy. NewsGenie puts those jobs in one chat window. It answers general questions, delivers category or topic news on demand, rates each story's source reliability, and remembers the conversation so follow-ups work naturally.

The system has four layers:

Layer	Technology	Job
Interface	Streamlit	Chat, category buttons, options, service status, per-session memory
Orchestration	LangGraph	Routes each message through classify → retrieve → filter → respond
Intelligence	OpenAI via LangChain	Intent classification (structured output) and grounded answers
Data	NewsAPI, GNews, DuckDuckGo	Real-time headlines and web search, in a fallback chain



NewsGenie interface with category news, reliability ratings, and workflow trace

2. AI chatbot design

2.1 Query differentiation

Every message is classified into one of five intents before any tool runs. Classification decides which APIs get called, so a general question never burns news API quota and a news request never gets a stale answer from the model's training data.

Intent	Example	Route
`greeting`	"hi", "what can you do"	Canned help text, no tools
`news_category`	"latest tech news", sidebar button	News providers by category
`news_topic`	"any news about Nvidia earnings?"	News providers by keyword + web background
`follow_up`	"tell me more about #2"	Remembered article + web search
`general`	"what's the difference between a stock and a bond?"	Web search + LLM answer

Classification runs in two layers:

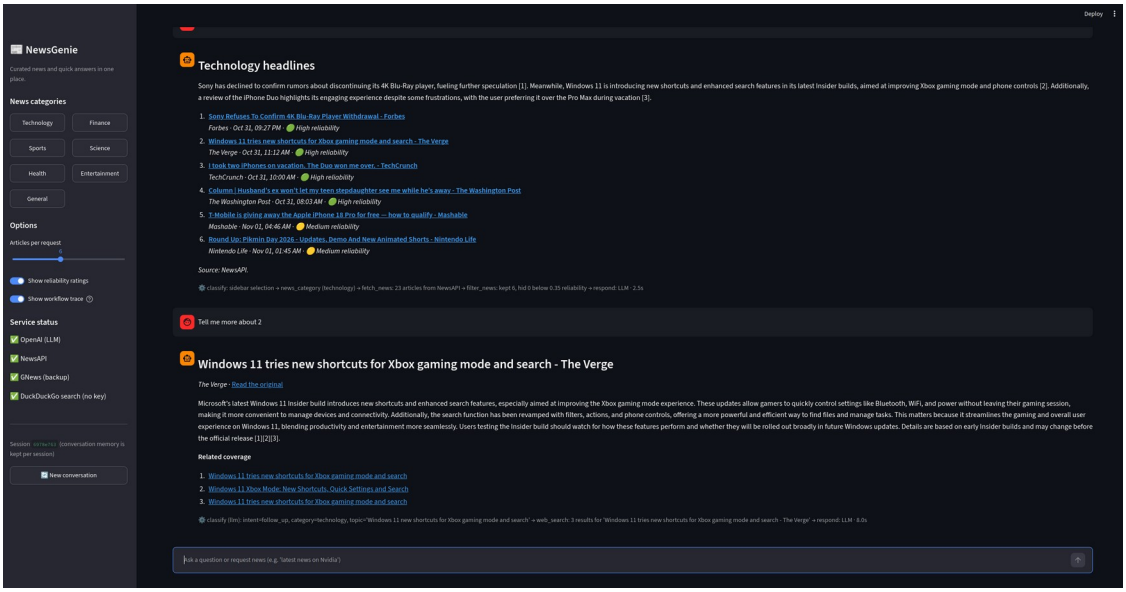
1. **Rules (always on, instant).** Regular expressions catch greetings, follow-up references ("#2", "the second article"), and news cue words ("latest", "headlines", "breaking"). Keyword lists map words to categories (for example "stocks", "Fed", "earnings" → finance). After removing filler words, anything left over becomes the topic, so "latest news on the Fed rate cut" yields topic *Fed rate cut*.
2. **LLM with structured output (when a key is set).** For anything the rules did not settle, the model returns a typed `Classification` object (intent, category, topic, article index, standalone search query). It sees the last six messages, which lets it resolve references like "what about him?" into a full search query.

Greetings and follow-ups skip the LLM entirely because the rules handle them reliably. If the LLM call fails or returns something inconsistent (a follow-up when no articles exist), the rule result is used. The UI trace shows which method decided.

2.2 Conversation management

- **Session isolation.** Each browser session gets a UUID `thread_id`. The compiled graph is shared across sessions, but LangGraph's checkpointer stores state per thread, so conversations never mix.
- **What persists between turns.** `messages` (full chat history, appended through the `add_messages` reducer) and `last_articles` (the most recent ranked list, which makes "tell me more about #3" work).
- **What resets each turn.** Intent, retrieved articles, web results, errors, and the trace are cleared in the first node so nothing from the previous request leaks into the next.
- **Context window control.** The classifier sees the last 6 messages and the answer generator the last 8. That keeps latency and cost flat as conversations grow.

- **New conversation.** A sidebar button issues a new `thread_id` and clears the visible chat.



Follow-up "Tell me more about 2" resolved by the LLM classifier using conversation memory

2.3 Response guidelines

- Ground answers in retrieved material and cite it as `[n]` with links.
- News briefings are 2 to 3 neutral sentences that only restate what the headlines say.
- General answers stay under about 200 words unless the user asks for more, and say so when no web results were available.
- Every news item shows source, time, and a reliability badge so users can judge for themselves.

2.4 Misinformation filtering

NewsGenie does not claim to fact-check. It applies a transparent, auditable reliability score to each article:

```
score = 0.55 × source reputation + 0.25 × headline quality + 0.20 × corroboration
```

- **Source reputation:** curated domain lists. Wire services and established outlets score 1.0, unknown domains 0.5, repeatedly low-rated sites 0.05, satire 0.0.
- **Headline quality:** penalties for clickbait patterns ("you won't believe", "!!!") and mostly-capitals headlines.
- **Corroboration:** credit when other independent outlets in the same batch carry a similar headline (word-overlap similarity).

Near-duplicate headlines (syndicated or lightly reworded copies of the same story) are collapsed into one entry, keeping the most reliable copy and crediting the others in its "also reported by" count. Stories below the threshold (default 0.35) and all satire are hidden from the

main list, but the UI lists them in an expander with the reason, so filtering is visible rather than silent.

3. Real-time news integration

3.1 Providers and categories

Order	Provider	Key needed	Notes
0	In-memory cache	No	10-minute TTL; repeated clicks are instant
1	NewsAPI	Yes	`top-headlines` for categories, `everything` for topics
2	GNews	Yes	Same interface, used if NewsAPI fails
3	DuckDuckGo News	No	Last live source; widens from 1 day to 1 week if empty
4	Last good result	No	Shown with a clear “cached copy from N min ago” warning

UI categories map to provider categories: Technology → `technology`, Finance → `business`, Sports → `sports`, plus Science, Health, Entertainment, and General.

3.2 Sample outputs

Technology

User: Show me the latest technology headlines

Workflow trace: classify: sidebar selection -> news_category (technology) → fetch_news: 23 articles from NewsAPI → filter_news: kept 6, hid 0 below 0.35 reliability → respond: LLM (2.6s)

NewsGenie:

Technology headlines

Sony has declined to confirm rumors about discontinuing its 4K Blu-Ray player, fueling further speculation [1]. Windows 11 is introducing new shortcuts and enhanced search features in its latest Insider builds, aimed at improving Xbox gaming mode and phone controls [2]. Meanwhile, a TechCrunch review highlights the engaging experience of using the iPhone Duo during a vacation, despite some frustrations [3].

1. **Sony Refuses To Confirm 4K Blu-Ray Player Withdrawal - Forbes**
   Forbes · Oct 31, 09:27 PM · 🟢 High reliability
2. **Windows 11 tries new shortcuts for Xbox gaming mode and search - The Verge**
   The Verge · Oct 31, 11:12 AM · 🟢 High reliability
3. **I took two iPhones on vacation. The Duo won me over. - TechCrunch**
   TechCrunch · Oct 31, 10:00 AM · 🟢 High reliability
4. **Column | Husband’s ex won’t let my teen stepdaughter see me while he’s away - The Washington Post**
   The Washington Post · Oct 31, 08:03 AM · 🟢 High reliability

5. **T-Mobile is giving away the Apple iPhone 18 Pro for free — how to qualify - Mashable**
   Mashable · Nov 01, 04:46 AM · 🟡 Medium reliability
6. **Round Up: Pikmin Day 2026 - Updates, Demo And New Animated Shorts - Nintendo Life**
   Nintendo Life · Nov 01, 01:45 AM · 🟡 Medium reliability

Source: NewsAPI.

Finance

User: Show me the latest finance headlines

Workflow trace: classify: sidebar selection -> news_category (business) → fetch_news: 21 articles from NewsAPI → filter_news: kept 6, hid 0 below 0.35 reliability → respond: LLM (1.4s)

NewsGenie:

Finance headlines

Binance has resumed withdrawals for some major cryptocurrencies in phases following a $6 billion heist, pending security checks [1]. Meanwhile, a family affected by Sanfilippo syndrome raised millions for gene therapy research, contributing to the FDA's recent approval of a related treatment [2]. In other news, bond investors are cautious ahead of the US midterm elections, and a judge has allowed a class-action lawsuit against McDonald's over its McRib sandwich to proceed [3][4].

1. **Binance resumes withdrawals for some major cryptocurrencies after $6 billion heist - CBS News**
   CBS News · Nov 01, 05:35 AM · 🟢 High reliability
2. **How one family gave a critical boost to gene therapy research - NPR**
   NPR · Nov 01, 05:00 AM · 🟢 High reliability
3. **Bond investors sit out 'most consequential' US midterm elections in years - Financial Times**
   Financial Times · Nov 01, 01:01 AM · 🟢 High reliability
4. **Judge won't dismiss McRib class-action lawsuit against McDonald's - USA Today**
   USA Today · Oct 31, 10:47 PM · 🟢 High reliability
5. **South Korea's Chip-Led Exports Accelerate, Bringing $1 Trln Annual Target Closer - WSJ**
   The Wall Street Journal · Oct 31, 09:42 PM · 🟢 High reliability
6. **We're Living in the Age of the Imperial CEO - WSJ**
   The Wall Street Journal · Oct 31, 09:00 PM · 🟢 High reliability

Source: NewsAPI.

Sports

User: Show me the latest sports headlines

Workflow trace: classify: sidebar selection -> news_category (sports) → fetch_news: 24 articles from NewsAPI → filter_news: kept 6, hid 0 below 0.35 reliability → respond: LLM (1.4s)

NewsGenie:

Sports headlines

Harriet Dart secured her first WTA title by defeating Lilli Tagger at the Guangzhou Open, marking a career highlight [1]. In college football, Week 9 saw notable performances with Nebraska and USC among the winners and losers, while projections for the College Football Playoff top 12 were updated [3][6]. Additionally, the Golden State Valkyries won the WNBA championship in their second season, led by Gabby Williams' 30 points in a close Game 7 victory over the Atlanta Dream [4].

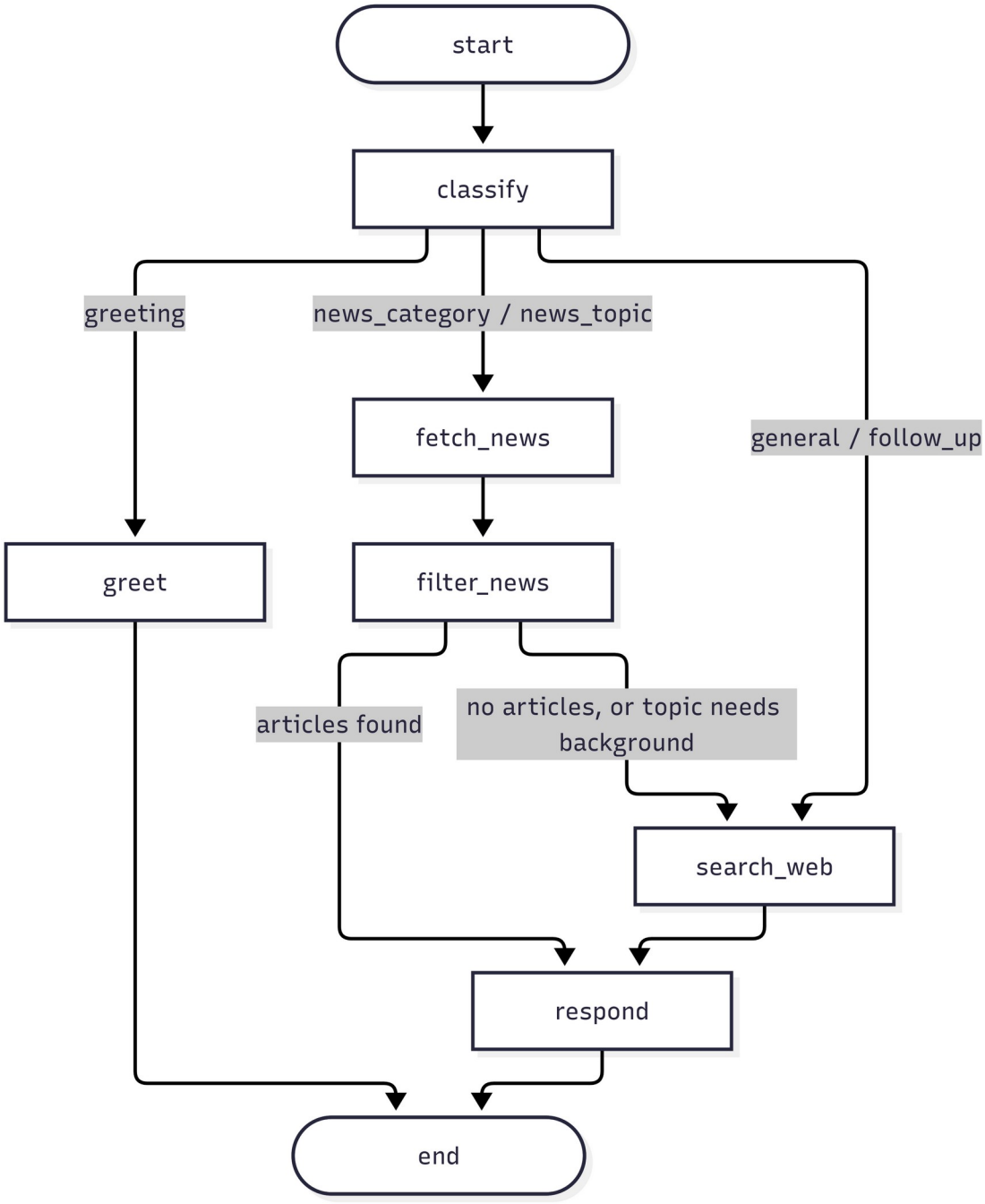
1. **Harriet Dart wins first WTA title by beating Lilli Tagger at Guangzhou Open - BBC**
   BBC News · Nov 01, 06:27 AM · 🟢 High reliability
2. **The Patriots are 18-43 all time in Miami. Here are keys to a Week 8 victory - ESPN**
   ESPN · Nov 01, 06:00 AM · 🟢 High reliability
3. **Nebraska, USC headline college football winners, losers in Week 9 - USA Today**
   USA Today · Nov 01, 05:07 AM · 🟢 High reliability
4. **Gabby Williams leads Valkyries to WNBA championship in second season, beating Dream 68-63 in Game 7 - AP News**
   Associated Press · Nov 01, 04:12 AM · 🟢 High reliability
5. **Fletcher's go-ahead score, Lightfoot's defensive play help No. 1 Miami rally past UNC 38-35 - AP News**
   Associated Press · Nov 01, 01:05 AM · 🟢 High reliability
6. **College Football Playoff top 12 projection after Week 9 - ESPN**
   ESPN · Nov 01, 12:25 AM · 🟢 High reliability

Source: NewsAPI.

`docs/failure_mode_demo.md` shows the same script run with every external service unreachable. The app still returned a clear message for each request instead of an error.

4. Workflow and error handling

4.1 LangGraph workflow



NewsGenie LangGraph workflow with routing conditions

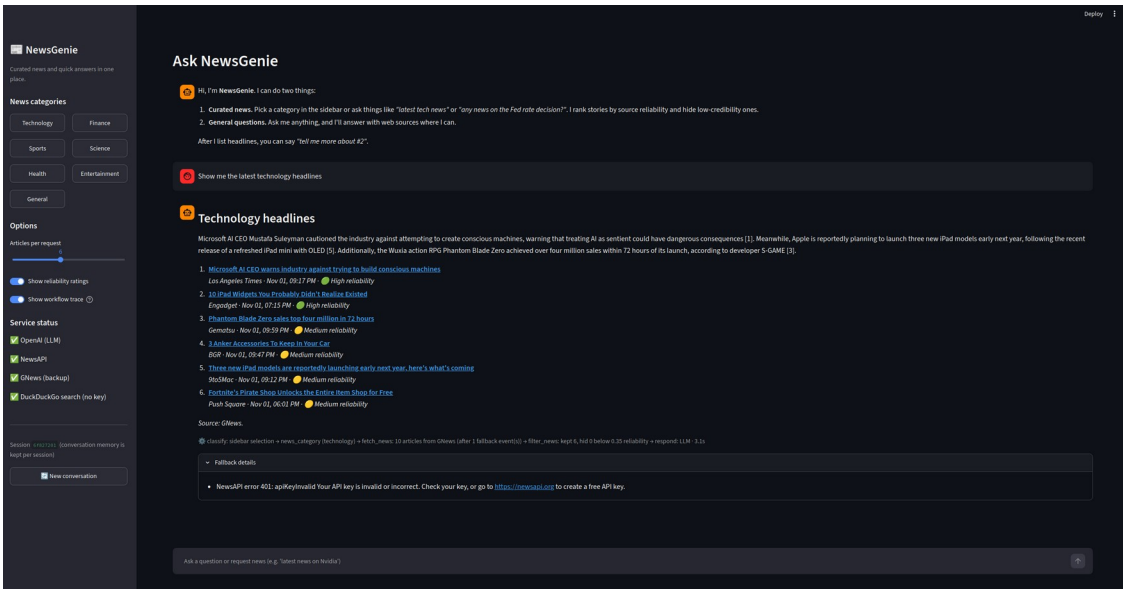
Node	What it does
`classify`	Resets per-turn state, records the user message, determines intent (or takes the sidebar category directly)
`greet`	Returns help text with no API calls

Node	What it does
`fetch_news`	Walks the provider chain and records every fallback event
`filter_news`	Scores, filters, and ranks articles; keeps the top N
`search_web`	DuckDuckGo search; query depends on intent (question, topic background, or article title)
`respond`	Builds the reply with the LLM, or with templates when the LLM is unavailable; saves `last_articles`

The two conditional edges are where fallback logic lives. After `filter_news`, an empty result sends the request to web search instead of ending with nothing.

4.2 Fallback matrix

Failure	Detected by	Behavior the user sees
`OPENAI_API_KEY` missing	`Settings.has_llm`	Rule-based routing and template replies; sidebar notice
LLM call errors or times out	try/except in classifier and responder	Rule classification / template reply for that turn; trace says "LLM failed"
`NEWSAPI_KEY` / `GNEWS_API_KEY` missing	Provider raises `ProviderError("... not set")`	Next provider in the chain; listed under "Fallback details"
HTTP 429 / 5xx	urllib3 `Retry` (2 retries, 0.5s backoff)	Retried automatically, then next provider
Timeout or network error	`requests` timeout (8s default)	Next provider
Provider returns no articles	Empty-list check	Next provider
Every live provider fails	End of chain	Last good result with stale warning, or web results, or a clear "try again" message
All articles fail reliability filter	`filter_news` returns empty	Routed to web search for alternatives
Follow-up number out of range	Bounds check	"I only have N articles from the last list"
Any unexpected exception	Top-level try/except in `app.py`	Friendly message; conversation is preserved



NewsAPI rejected an invalid key (401); the workflow fell back to GNews and still delivered headlines

4.3 Performance

- **Caching.** News results cache for 10 minutes per category or topic, and web results for 15 minutes, which cuts both latency and API quota use.
- **Skipping work.** Greetings and sidebar category clicks skip LLM classification. Category news skips web search unless news retrieval came back empty.
- **Bounded context.** History is truncated before LLM calls.
- **Shared resources.** The graph and HTTP session are built once per server process (`st.cache_resource`) instead of on every rerun.
- **Visibility.** Each reply shows the nodes that ran, the provider used, and elapsed seconds.

4.4 Testing

The `tests/` folder has 32 pytest tests that run offline by replacing network calls with fakes. They cover intent classification, topic extraction, reliability scoring, every step of the provider fallback chain (including stale cache and total failure), session isolation, follow-up memory, and the LLM path through a stub model.

5. Limitations and next steps

- The reliability score is a heuristic. A next step is pulling ratings from a published media-bias dataset instead of hand-maintained lists, and adding claim-level checks against fact-check APIs.
- NewsAPI's free plan only works from localhost, so a public deployment would need a paid plan or GNews as primary.
- NewsAPI assigns categories by outlet rather than by article, so category feeds occasionally include off-topic stories (for example, an advice column under Technology). A per-headline relevance check would tighten this.

- `InMemorySaver` loses history when the server restarts. A SQLite or Postgres checkpointer would make sessions durable.
- Personalization (saved favorite categories, topic alerts) would build naturally on the existing session state.

References

GNews. (n.d.). *GNews API documentation (v4)*. Retrieved October 28, 2026, from https://gnews.io/docs/v4

LangChain. (n.d.). *LangGraph documentation*. Retrieved October 28, 2026, from https://langchain-ai.github.io/langgraph/

NewsAPI. (n.d.). *Documentation*. Retrieved October 28, 2026, from https://newsapi.org/docs

Streamlit. (n.d.). *Streamlit documentation*. Retrieved October 28, 2026, from https://docs.streamlit.io/